# Coordinated Relational Evolution for Zero-Day Intrusion Detection on Temporal Network Graphs

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**Abstract**

Zero-day intrusion detection is difficult due to the fact that previously unknown attacks can display local behavior that fits the existing network while changing the arrangement of communication relationships over time. The existing intrusion detection methods based on temporal graphs are focused mainly on representing the changing network state and discovering unusual patterns based on the representation of nodes and edges. In this paper, we focus on the relational-transition approach that considers the changes in the communications taking place in consecutive communications explicitly. We create causal temporal communication graphs based on network flows and derive the processes of behavioral transitions of stable source-and-destination communications using traffic statistics. The transitions are then studied concerning their relational organization throughout each time window under the strict normal-only training approach with the Shellcode attack family excluded from the training. The CTGAD-X algorithm shows ROC-AUC equal to 0.9771 and PR-AUC equal to 0.9898 and during the relational analysis, it is found out that there is a huge difference in transition organization between normal windows and windows with attacks present: the standard deviation of transition degree is 0.3034 for normal windows and 4.1694 for windows with attacks.

**Keywords:** zero-day intrusion detection, temporal graphs, relational evolution, graph anomaly detection, network security, behavioral transitions

# 1 Introduction

Detecting zero-day intrusions is difficult because the distinguishing features of a new attack cannot be obtained in the training process. This implies that the detection system should be able to determine the anomalies of the behavior without the help of the knowledge about the attack in question. In the case of network traffic, this problem can be represented in terms of time and relational representation because flows of communications connect various hosts, services, and endpoints, while their behavior changes over time.

As a result, there has been a growing interest in developing intrusion detection systems based on graphs, since the components of the network can be represented in the form of nodes, while communication activity can be illustrated as edges [1]. Moreover, the temporal graph formulation allows to take into account the changes in network traffic across various time periods [2–4]. However, having information about the states of a network in different time intervals does not guarantee information about how the relationships change over time. In fact, similar aggregated statistics can show different types of changes in simultaneous relationships.

The paper investigates the issue mentioned above in a relational-transition framework, which implies that apart from observing the states of temporal communication graphs, the paper tracks the change in the behavior of continuous source-destination relationships between adjacent time windows. Flow-level statistics are used to represent the existing relationship, and their transition is understood through the variations in those statistics. The transitions that happen on the same time span get organized by the nature of their relational interactions. In this paper, we proceed from the idea that malicious actors may not just cause the appearance of isolated anomalous edges in the communication network but may trigger synchronized changes in more than one communication relationship. Such synchronization is likely to speak in favor of structural proofs that would complement the traditional temporal embeddings of anomaly detection. The authors rely on the CTGAD-X approach for the analysis and carry out the analysis under a pure normal-only training regime with the Shellcode family excluded from the training process [5]. The analysis results in a baseline that can be used as a meaningful benchmark for the assessment of the performance of other temporal and relational approaches.

This contribution of our research comprises:

- We present a relational-transition representation of temporal network data aimed at modeling the changes of persistent relationships observed in successive snapshots of the graph.
- We develop a representation of interaction aimed at considering simultaneous relationship transitions, to be able to study how the relationships change, not only indicating deviations at the level of edge transitions within the system.
- We undertake an evaluation without prior knowledge of the Shellcode attack family in the process of training and compare the obtained relational analysis to various processes of temporal embedding and ablation.

- We examine empirically how the relations change in normal and attack traffic environments, as well as the discrepancy in transition-level changes and interaction volume.

The rest of this paper is arranged as follows. Section II describes previous research. Section III establishes the research problem. Section IV presents the methodology. Section V explains the experimental setup. Section VI conveys the findings and analysis, followed by Section VII discussion. Finally, Section VIII provides conclusions.

# 2 Related Work

## 2.1 Temporal Graph-Based Intrusion Detection

More recently graph neural networks have been more frequently used. in network intrusion detection as communication relationships can be easily expressed in graphs [1, 6]. Temporal graph models build upon this idea by processing successive graphs and capturing the evolvement of network behavior [2–4, 7]. Nevertheless, temporal graph approach usually focuses on how the network state is represented at the particular moment and how it changed in the latent space. This leads to another question, namely whether change structure in communication relationship contains information about the network behavior in case of attack.

## 2.2 Relational and Structural Evolution

Detecting anomalies in dynamic graphs refers to the process wherein the evolution of structures, shifting of relations, and higher-level interactions is taken into account. In these approaches, the argument is made that anomalies do not necessarily have to concern separate nodes or edges but that shifting relationships can also serve as a good indicator of the occurrence of an anomaly.

In terms of network security, this viewpoint is important since coordinated attacks could change numerous relationships within a limited time period. However, the need to model changes in relations precisely means that one has to figure out the correct behavior state for each relationship and differentiate between normal traffic development and coordinated structural change.

## 2.3 Zero-Day and Explainable Detection

Zero-day detection often depends on learning representations of normal behavior or previously seen behavior and looking for anomalies in the inference process. The normal-only learning approach is especially useful when there are no attack labels available for new threats. Explainability is important in operational security as anomaly scores do not give any useful information as to why the traffic pattern in question is marked as suspicious.

Thus, this paper follows the temporal graph anomaly detection framework but also studies whether the use of explicit relational transitions gives any further behavioral insight. The aim is not to substitute temporal graph models but rather to assess if

the organization of relational changes can provide additional evidence for zero-day detection.

# 3 Problem Formulation

Let a network traffic stream be divided into a sequence of causal temporal snapshots

$$\mathcal{G} = \{G_1, G_2, \ldots, G_T\}, \tag{1}$$

where each snapshot is represented as

$$G_t = (V_t, E_t, X_t). \tag{2}$$

Here, $V_t$ denotes network entities, $E_t$ denotes directed communication relationships, and $X_t$ represents node-level behavioral features.

For a persistent communication relationship between source $u$ and destination $v$, let its behavioral state at time $t$ be

$$r_{uv}^t = [r_{uv,1}^t, r_{uv,2}^t, \ldots, r_{uv,d}^t], \tag{3}$$

where the components describe traffic characteristics such as flow count, bytes, packets, duration, load, and TTL statistics.

The relational transition between two consecutive snapshots is defined as

$$\Delta r_{uv}^t = \phi(r_{uv}^{t-1}, r_{uv}^t), \tag{4}$$

where $\phi(\cdot)$ represents the transition operator. In our implementation, the transition for each behavioral dimension is computed using a signed logarithmic change:

$$\Delta r_{uv,j}^t = \log(1 + r_{uv,j}^t) - \log(1 + r_{uv,j}^{t-1}). \tag{5}$$

The magnitude of a relational transition is then defined as

$$m_{uv}^t = \left\| \Delta r_{uv}^t \right\|_2. \tag{6}$$

For each temporal interval, the resulting set of persistent relational transitions is

$$\mathcal{T}_t = \{(u, v, \Delta r_{uv}^t, m_{uv}^t)\}. \tag{7}$$

Instead of looking at each transition separately, we take the next step to show how transitions within the same time span are connected through communicating points and arrangement of events. To accomplish this, we create a transition level relational structure that represents how simultaneous behavioral changes occur.

# 4 Methodology

Figure 1 outlines the entire analysis process. Flow of the networks is first transformed into temporal causal communication graphs. Then, snapshots obtained would be processed through two different perspectives: CTGAD-X temporal encoder and the provided relational-transition analysis. In turn, the first, produces normal-reference embedding for the zero-day anomaly detection; while the latter describes the manner how behavior deviations evolved across stable communication links throughout the time interval.

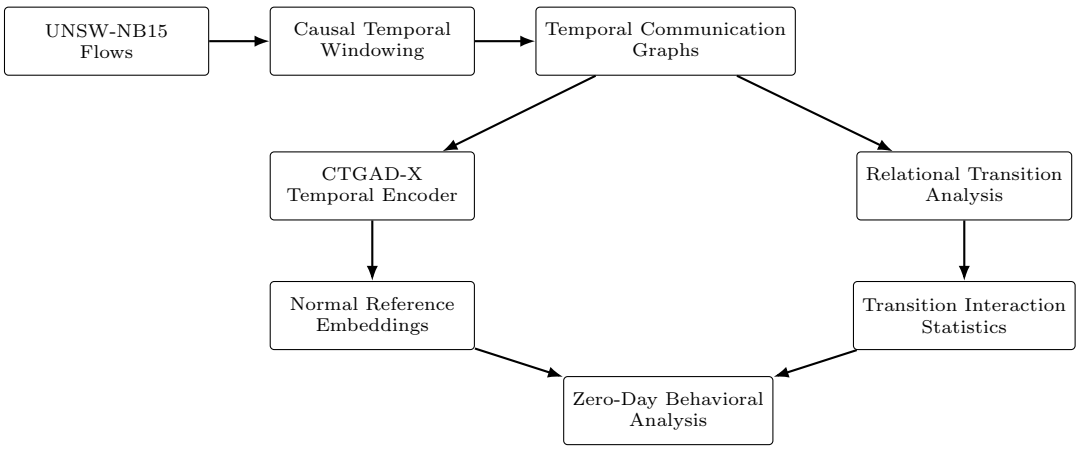



**Fig. 1** Overall framework for temporal graph and relational-transition analysis.

## 4.1 Causal Temporal Graph Construction

The network traffic is converted into causal temporal snapshots by using fixed time slots. A specific time slot obtains the flow traffic statistics depending on the flow's involvement in that time window. The network traffic of the flow is calculated by using these rate expressions, while the statistics of TTL is obtained from the respective temporal clocks of the flow.

This procedure ensured the causality of time and creates the series of communication graphs, which may be used for performing temporal embedding and relational transition analytics.

## 4.2 CTGAD-X Baseline

A graph-temporal approach is utilized in the baseline, wherein each snapshot of the graph is processed via a graph attention network [8] . It extracts node embeddings via mean-pooling to create the snapshot representation. This creates a set of representations for the snapshots, which go through temporal multi-head attention to get a temporal embedding [9]. The encoder is optimized via normal traffic data and a contrastive loss during the training phase [10]. The projection layer is only utilized during contrastive training. When inference occurs, the encoder is used to create temporal embeddings for anomaly detection, but the projection layer is not used.

## 4.3 Relational Transition Representation

A relational transition is generated only when a source–destination relationship persists across consecutive temporal windows. Figure 2 illustrates the formulation. The relationship state changes from $r_{uv}^{t-1}$ to $r_{uv}^{t}$, producing a vector-valued transition $\Delta r_{uv}^{t}$. Its magnitude summarizes the amount of behavioral change, while the collection of transitions within the same temporal interval is used to characterize their interaction structure.

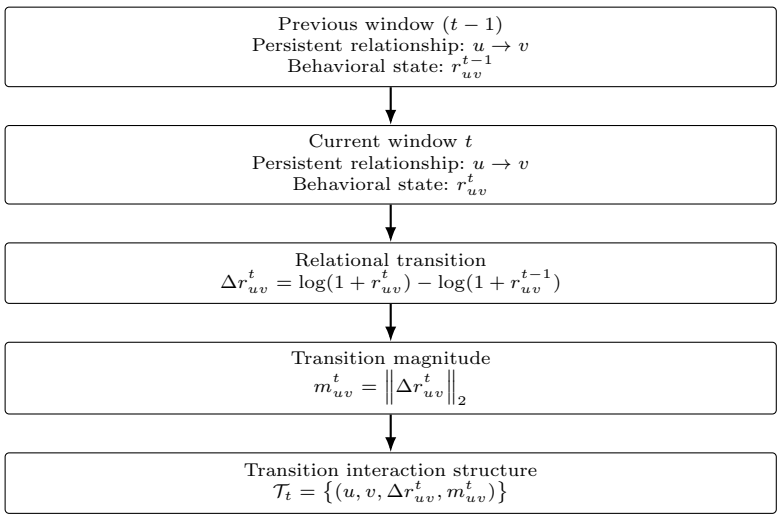



**Fig. 2** Construction of a relational transition from consecutive relationship states.

The analysis proposed works with constant relationships between sources and destinations. For each relationship that appears in consecutive windows, we create a relational state based on aggregating flow-level statistics. The logarithmic formulation reduces the impact of large traffic values, but we still manage to preserve the direction of the behavioral change.

The transition magnitude $m_{uv}^{t}$ provides a compact measure of how strongly a relationship changes. Importantly, a large transition magnitude alone is not treated as evidence of malicious activity because substantial behavioral changes can also occur during legitimate network evolution.

## 4.4 Transition Interaction Structure

In order to examine the collaborative change of relationships, individual transitioning events are seen as components of the transition interaction structure. When two transition events use the same communication terminals or are sequenced, an interaction occurs. This gives the transition interaction structure such parameters as interaction density, mean transition degree, and degree variability that can be calculated.

For a transition interaction graph with $N_t$ transition nodes and $E_t^{\text{int}}$ interaction edges, its interaction density is computed as

$$D_t = \frac{2E_t^{\text{int}}}{N_t(N_t - 1)}. \tag{8}$$

Variation in degree is assessed through the standard deviation of the degrees of degree of transition nodes. These statistics are employed as descriptive indicators of how relational transformation is organized instead of being used as detached classification scores.

**Table 1** Dataset and Temporal Split

| Property | Value |
|---|---|
| Total flows | 200,000 |
| Normal flows | 177,785 |
| Attack flows | 22,215 |
| Total attributes | 52 |
| Temporal windows | 255 |
| Training windows | 0–177 |
| Validation windows | 178–203 |
| Test windows | 204–254 |
| Held-out attack family | Shellcode |
| Test normal sequences | 15 |
| Test attack sequences | 32 |

# 5 Experimental Setup

## 5.1 Dataset

Studies are conducted utilizing the UNSW-NB15 dataset that involves network intrusions [11]. The dataset comprises around two hundred thousand traffic instances represented by a set of 52 attributes. Among these instances 177,785 were normal traffic instances, while 22,215 instances were traffic detalis about attacks and attacks consist of different traffic types including Generic, Exploits, Fuzzers, Reconnaissance, DoS, Backdoors, Analysis, Shellcode and Worms [12, 13].

The traffic streaming is temporally ordered and divided into 255 causal windows. The experimental procedure relies on defining causal windows chronologically instead of randomly dividing traffic instances in order to eliminate the influence of future traffic on the obtained results regarding past observations.

## 5.2 Zero-Day Evaluation Protocol

The methodology of the principal evaluation is designed in such a way so that the traffic of Shellcode is not used for the model training but is still applicable for the testing and evaluation time. In other words, at the training phase, we use solely normal traffic data which accumulates into 178 normal-only temporal snapshots and 174 normal training sequences.

The normal reference bank consists of 174 frozen embeddings of the temporal data created by the trained encoder. The test data undergoes temporal encoding and is then compared to the normal reference. In doing so, the method ensures that the attack family was not involved in the generation of the normal set.

The test data contain 47 temporal sequences of which 15 are normal sequences.

**Table 2** Shellcode-Held-Out Zero-Day Detection

| Method | ROC-AUC | PR-AUC |
|---|---|---|
| CTGAD-X | 0.9771 | 0.9898 |

### 5.3 Training Configuration

The CTGAD-X encoder operates utilizing an input dimension of 11, hidden representation size 60, 4 structural and 4 temporal attention heads, with the temporal sequence of 5 snapshots. The contrastive projection head then values the 60-dimensional encoder representation in passing it through a hidden layer with 60 dimensions to attain a projection space of 80 dimensions.

This encoder features a total of 20,820 trainable parameters, while its projection head encompasses 8,540 parameters. The contrastive training lasts for 100 epochs, with the best validation occurring at epoch 80 with the contrastive loss of 0.00295. After the training, the projection head is abandoned and does not partake in any anomaly scoring processes.

### 5.4 Evaluation Metrics

Receiver operating characteristic area under the curve (ROC-AUC) and precision-recall area under the curve (PR-AUC) measures the performance of the algorithms. While ROC-AUC is an indicator of how well the classification performs for various threshold values, PR-AUC gives more insight about the quality of classification in cases where the normal and outlier instances are unbalanced.

In associative transition analysis, we also present the number of transitions, interaction density, average transition degree, and standard deviation of the degree. These metrics describe the structure of relational changes, as opposed to substituting the baseline anomaly score.

## 6 Results and Analysis

### 6.1 Zero-Day Detection Performance

According to the evaluation carried out by Shellcode, the CTGAD-X baseline scored a ROC-AUC of 0.9771 and a PR-AUC of 0.9898. The results provide a robust temporal representation against which other behavioral representations can be tested.

This finding is crucial for future relational analysis as it indicates that the basic representation of the temporal graph can offer a lot of data concerning the previously unknown techniques of the attack. This underlines the importance of relational-transition analysis not in proving that existing representations are inadequate, but in checking whether there exist different methods for presenting the change in time.

**Table 3** Ablation Results Under Shellcode-Held-Out Evaluation

| Variant | ROC-AUC | PR-AUC |
|---|---|---|
| CTGAD-X | 0.977083 | 0.989811 |
| NKVEL: Raw Temporal Evolution | 0.664368 | 0.851568 |
| SAGE | 0.978495 | 0.989754 |
| CTC | 0.503333 | 0.679726 |
| State + CTC | 0.666667 | 0.832702 |
| Conformal Calibration | 0.977083 | 0.989870 |
| TBLA | 0.719792 | 0.870559 |

## 6.2 Ablation Study

Several alternative representations were evaluated under the same held-out Shellcode protocol. The results are summarized in Table 3.

The performance of raw temporal evolution rendering is significantly inferior to that of CTGAD-X because ROC-AUC drops to 0.6644. This shows that measuring the motion of latent representations alone does not provide a proper characterization of the unseen attack behavior.

The SAGE version produces a ROC-AUC of 0.9785 and is similar to the baseline, at the same time, showing pretty much the same value of PR-AUC. In the same way, conformal calibration affects PR-AUC only slightly so that the numbers actually change from 0.989811 to 0.989870 while keeping ROC-AUC intact. Such results point out to the fact that neither of the techniques has a great impact on ranking behavior.

CTC-based models perform much worse than the CTGAD-X system. The CTC-based representation alone leads to a ROC-AUC of 0.5033, while adding state representation improves this figure to 0.6667, still leaving behind the benchmark.

TBLA which describes the trajectories of the behavior achieved ROC-AUC of 0.7198, and PR-AUC of 0.8706 which means that working with trajectories of latent variables is not enough to understand the structure of interrelations in the network.

## 6.3 Relational Transition Analysis

Relational-transition modeling offers a new perspective on the functioning of temporal networks. Throughout the duration of a test held aside, the number of persistent relational transitions reached 5,316. The behavioral state of each source-destination connection was established and compared across different time windows.

The results of transition interactions being recorded were classified under attack and non-attack classifications. The general statistics are presented in the table 4.

The distinctive feature is that the transition degree variance differs greatly. The standard deviation of transition degree in normal windows is 0.3034 while in the case of windows with attacks, it reaches 4.1694. Therefore, the variability of transition degree in these windows is around fourteen times higher as opposed to normal windows.

Interaction density goes down from 0.1818 to 0.1495 when we compare normal windows and windows with attacks. At the same time, there are more transition nodes on average in the attack windows. The data confirms that the difference between normal and attack periods is more than simply having more transitions.

**Table 4** Relational Transition Organization in Test Windows

| Statistic | Normal | Attack |
|---|---|---|
| Transition nodes/window | 99.00 | 110.81 |
| Interaction edges/window | 882.21 | 906.74 |
| Interaction density | 0.1818 | 0.1495 |
| Mean transition degree | 17.82 | 16.41 |
| Degree standard deviation | 0.30 | 4.17 |

All this matters because the occurrence of single large change is not a synonym for malicious actions. Network activities can result in considerable changes in the volume of traffic or other flow statistics too. Thus, instead of comparing the amounts of transitions that happen we deal with whether the relationships that are changing remain homogeneous.

### 6.4 Interpretation

The research results yield a basis for distinguishing between two types of activity. One of them is behavior volume, which determines the degree of changes in any specific interaction process. The second one is behavior structure, which identifies how transformations within various activities take place at the same timeframe.

The CTGAD-X findings indicate time-based graph representations can result in effective zero-day attack identification [14]. Moreover, the results reveal that replacing this form of representation with raw latent evolution and trajectory-based representation versions negatively affects the quality of detection.

Relational transition statistics provide evidence. During non-attack occurrences, the transitions degrees do not vary a lot. On the contrary, in non-normal conditions, the degrees of transition may show significant variability and lower collaboration density. This proves that changes in relational behavior can reflect important structural features that cannot be revealed by single edge transition volume.

It is necessary to mention that these statistics do not suppose independent training of an attack classifier. Their main function is to characterize the structure of timebased behavior and to justify using relational transition statistics.

## 7 Discussion

The results point out a key factor for intrusion detection within temporal graphs. A network can face many legitimate changes during its functioning without being classified anomalously on a global scale. In this way, it appears that using only the value of certain changes for anomaly detection may result in confusion between the normal evolution of the situation and malicious activity.

This is the point of view provided by relational transition analysis. Instead of simply asking whether or not a communication link changes, this approach considers how different changing links are placed in time. The significantly greater variability level in

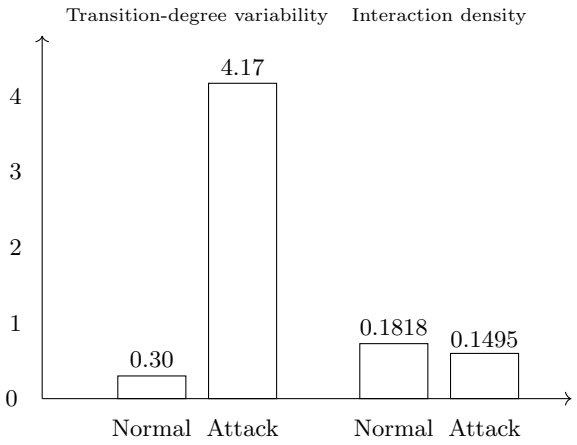



**Fig. 3** Relational organization of normal and attack-containing test windows.

the windows related to attacks indicates that the attacks are probably distinguished by a less homogeneous organization of transitions.

Nonetheless, this should be approached carefully. The current analysis shows there is an empirical structural distinction between the datasets evaluated in this study, but it does not mean that the statistics used are universally distinguishing for different datasets and network environments.

Notwithstanding that, the findings imply that temporal network anomaly detection method can be enhanced not only by taking into account configurations of links in the graph but also by applying some other approaches as well. Thus, it can be said that the emphasized representation contributes to another type of revealing the behavior that can be used for reinforcing the previous cases.

# 8 Conclusion

This research introduces the idea of using coordinated relational evolution as a different approach for detecting zero-day attacks on temporal network graphs. We started from a basic normal-only CTGAD-X setup and represented consistent relationships that exist between two endpoints with behavioral states that we later examined.

In this study, using the method of Shellcode-held out evaluation CTGAD-X showed a value of ROC-AUC equal to 0.9771, while the PR-AUC was equal to 0.9898 With the introduction of alternative latent-evolution and trajectory-based concepts the achieved results could not be explained solely by movement in representation.

When using relational-transition method the analysis of transition degree was made and two groups of data were analyzed. The first group of data contained data on the detected attacks and showed higher degree of transition degree variable along with lower interaction density than that in the second group of data.

The analysis of future research will include the idea of direct learning via relational transitions and capabilities of being used in the future research.

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